Month 1 2 3 4 5 6 7 8 9 10 11 12 Date 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31

LLM 5-dars

- Block foundations
- Transformer
- Training and Performance.

Block Foundation:

- Layer ⇒ Neyron tarmoq kabi qatlam inputdan outputgacha

LLM NN Layerlar 2 qismdan iborat ekan:

- Attention layer → tokenlar orasidagi bog'liklik
- Feed-forward Network (FFN)

Fully Connected Layers

↓ 2ta fully connected layer bor

first model expands the input dimension dmodel from 512 to 2048, 2nd one projects it back to 512

→ Activation function: ReLU non-linearity to learn more complex patterns

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Position-wise processing: Positionga qarab ōrganish ekan.

$$FNN(x) = \max(0, xW1 + b1)W_2 + b2$$

Attention layer: Tokenlar orasidagi bog'likni ōrganadi.

FNN -> mo'no xosil qiladi, bilim xosil qiladi

Har Bir tokenni alohida ishlaydi

GQA. SwiGLU / GeGLU

- Serial: Data -> Attention -> FNN

- Paralel: Data ↗ Attention ↘ ↘ FNN ↗ -> Birlashish

  -> Tezroq, same smartness, sanoat darajasi

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Normalization: → RMSNorm → zamonaviy usul

↙

Layer Normalization ⇒ Mean, Variance

↙

Eski usul

RMSNorm ⇒ GPU ni 10~50% gacha qisqartiradi shu sababli tezroq ishlaydi.

Vocabulary and Tokenization.

Tokenization ⇒ Breaks sentence/words into chunks.

Vocabulary ⇒ Chunklarni öz ichki röyxatidan qidiradi

Pre-Norm: Input -> Tokenization -> Normalization -> Attention/FFN -> Activation

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GPU out of Memory ⇄ yechim

Attention Bilan hal qilinar ekan GQE

$\boxed{\text{Block foundation} \quad \text{Attention/FFN} \Rightarrow \text{Paralel}}$

↓

Normalization -> RMSNorm

↓

Activation -> SWIGLU / GeGLU

↓

$\boxed{\text{Transformer Architecture}}$

~~Vo~~ Tokenization ⇄ Vocabulary

↓

$\boxed{\text{Pre-Norm}}$